CERTIFIED ETHICAL HACKER | SCANNING AND ENUMERATION

# Enumeration

## Concepts, Techniques, Tools, Lab Walkthroughs and Countermeasures

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| 1 | Enumeration concepts |
| 2-5 | NetBIOS, SNMP, LDAP and NFS enumeration |
| 6-7 | SMTP and DNS enumeration; other enumeration techniques |
| 8 | Enumeration countermeasures |
| 9 | Hands-on labs: Windows CLI, NetBIOS Enumerator, NSE, snmp-check, Active Directory Explorer, RPCScan, Global Network Inventory |
| 10-11 | Quick revision sheet and practice questions |

**Important:** Enumeration sends direct queries to live systems. Perform every technique in this guide only on systems you own or have **written authorization** to test (for example a CEH lab or your own virtual lab). Unauthorized enumeration may be illegal.

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# 1. Enumeration Concepts

## What is enumeration?

Enumeration is the process of **actively connecting to a target and extracting detailed information** from it: usernames, machine names, network resources and shares, services, banners, routing tables, and similar data. It comes after footprinting and scanning, once you know which hosts are alive and which ports are open. Unlike scanning, enumeration establishes an **active connection** and issues directed queries, so it is more intrusive and more likely to be logged.

## Information gathered

- Usernames, group names and group memberships
- Hostnames, NetBIOS names and domain/workgroup names
- Network shares and exported file systems
- Running services, service banners and OS details
- IP tables, routing tables and SNMP data
- Application names and versions; password policy details; DNS and mail records

## Why it matters

The data collected lets an attacker (or a penetration tester) identify weak points such as valid accounts for password attacks, open shares, misconfigured services and outdated software. For defenders, knowing what is exposed is the first step to closing it.

## Techniques for enumeration

| Technique | Description |
|---|---|
| Extract usernames using email IDs | The part before @ in an email address is often the user name (e.g. jsmith@corp.com gives jsmith). |
| Extract information using default passwords | Many devices and applications ship with default credentials that are never changed. |
| Brute-force Active Directory | Exploits weak or poorly managed AD authentication to discover valid accounts. |
| Extract usernames via SNMP | Poorly secured community strings expose user and device data. |
| Extract information via DNS zone transfer | A permissive name server hands over its full zone: hosts, IPs, services. |
| Extract user groups from Windows | Using built-in tools or null/guest sessions to list groups and members. |

## Services and ports commonly enumerated

| Service | Port(s) | Protocol | What can be learned |
|---|---|---|---|
| NetBIOS Name Service | 137 | UDP | Host names, domain, logged-on users |
| NetBIOS Datagram / Session | 138 / 139 | UDP / TCP | Share and session information |
| SMB (direct hosting) | 445 | TCP | Shares, users, OS, policies |
| SNMP | 161 / 162 | UDP | Devices, interfaces, processes, users |
| LDAP / LDAPS | 389 / 636 | TCP | Directory objects: users, groups, computers |

| Service | Port(s) | Protocol | What can be learned |
|---|---|---|---|
| Global Catalog | 3268 / 3269 | TCP | Forest-wide AD search |
| NFS | 2049 | TCP/UDP | Exported file systems |
| RPC portmapper | 111 | TCP/UDP | Registered RPC services |
| SMTP | 25 (also 465/587) | TCP | Valid mailbox users |
| DNS | 53 | UDP/TCP | Hosts, records, zone data |
| NTP | 123 | UDP | Hosts, clients, system details |
| Telnet / SSH | 23 / 22 | TCP | Banners, version info |
| FTP / TFTP | 21 / 69 | TCP / UDP | Banners, anonymous access, files |
| SIP (VoIP) | 5060 | UDP/TCP | Extensions, VoIP users |
| IPsec / IKE | 500 | UDP | VPN gateway presence and configuration |

Memorise this table: port-to-service mapping is a very common exam question.

# 2. NetBIOS Enumeration

## NetBIOS basics

**NetBIOS** (Network Basic Input/Output System) lets applications on different computers communicate over a LAN. Each Windows machine has a NetBIOS name of up to **16 characters**: 15 characters for the name and a final 16th character, the **suffix**, that identifies the service. NetBIOS runs over TCP/IP (NBT) using ports 137 (name service), 138 (datagram) and 139 (session).

If port 139 (or 445) is open and file and printer sharing is enabled, an attacker can query NetBIOS name tables to learn computer names, domain/workgroup names, MAC addresses, logged-on users and shares.

## NetBIOS name codes (suffixes)

| Code (hex) | Type | Meaning |
|---|---|---|
| <00> | UNIQUE | Workstation service (computer name) |
| <00> | GROUP | Domain or workgroup name |
| <03> | UNIQUE | Messenger service (often reveals logged-on user name) |
| <20> | UNIQUE | File Server service (file sharing is enabled) |
| <1B> | UNIQUE | Domain Master Browser |
| <1C> | GROUP | Domain Controllers |
| <1D> | UNIQUE | Master Browser |
| <1E> | GROUP | Browser service elections |
| <BE> | UNIQUE | Network Monitor agent |

## Tools and approach

- **nbtstat** (built into Windows): queries NetBIOS name tables and caches.

- **net view / net use**: lists shares and connects to them.
- **NetBIOS Enumerator** (GUI): scans an IP range for NetBIOS information.
- **Nmap NSE** scripts: nbstat, smb-os-discovery, smb-enum-shares, smb-enum-users.
- Others: Hyena, SuperScan, NetScanTools Pro, Nsauditor, nbtscan, enum4linux.

### Key nbtstat switches

| Command | Purpose |
|---|---|
| nbtstat -a <hostname> | Name table of a remote machine by NetBIOS name |
| nbtstat -A <IP> | Name table of a remote machine by IP address |
| nbtstat -c | Contents of the local NetBIOS name cache |
| nbtstat -n | Local NetBIOS names |
| nbtstat -S | Sessions table (with IP addresses) |
| nbtstat -r | Names resolved by broadcast and WINS |
| nbtstat -R | Purge and reload the remote cache name table |

# 3. SNMP Enumeration

## What is SNMP?

**Simple Network Management Protocol** is used to monitor and manage network devices such as routers, switches, servers, printers and workstations. It runs over UDP: **161** for queries and **162** for traps (notifications from agents).

## Components

| Component | Role |
|---|---|
| Managed device | Any node with an SNMP agent (router, switch, server, printer). |
| Agent | Software on the device that stores data and answers requests. |
| NMS (manager) | Management station that polls agents and receives traps. |
| MIB | Management Information Base: a hierarchical, virtual database of manageable objects. |
| OID | Object Identifier: the dotted numeric address of an object in the MIB tree. |

## Community strings and versions

SNMPv1 and v2c authenticate with **community strings** sent in clear text. The default **read-only** string is typically **public**, and the default **read-write** string is typically **private**. If defaults are left in place, anyone can read, and possibly modify, device configuration.

| Version | Authentication / security |
|---|---|
| SNMPv1 | Community string in clear text; no encryption. |
| SNMPv2c | Community string in clear text; adds GETBULK and better error handling. |
| SNMPv3 | Username-based, supports authentication (MD5/SHA) and encryption (DES/AES). Recommended. |

### SNMP operations

- **GetRequest / GetNextRequest / GetBulkRequest**: read values from the agent.
- **SetRequest**: change values on the agent (needs write access).
- **Response**: the agent's reply. **Trap / InformRequest**: unsolicited alerts from agent to manager.

### Information exposed

Network interfaces and IP addresses, routing tables, ARP cache, running processes, installed software, listening ports, user accounts, system uptime, hostnames and device descriptions.

#### Common OIDs on Windows (LanMan MIB)

| OID | Information |
|---|---|
| 1.3.6.1.4.1.77.1.2.25 | User accounts |
| 1.3.6.1.4.1.77.1.2.3.1.1 | Running services |
| 1.3.6.1.4.1.77.1.4.1 | Domain name |
| 1.3.6.1.4.1.77.1.2.27 | Share names |

Tools: snmp-check, snmpwalk, snmpget, Nmap snmp-* scripts, SoftPerfect Network Scanner, OpUtils, Engineer's Toolset, Net-SNMP.

# 4. LDAP Enumeration

## What is LDAP?

**Lightweight Directory Access Protocol** is an Internet protocol for accessing and maintaining distributed directory services. Microsoft Active Directory (AD), OpenLDAP and similar products use it. Clients send queries to a Directory System Agent (DSA) on **TCP 389** (clear text or STARTTLS), **636** (LDAPS), and the Global Catalog on **3268 / 3269**.

## Directory structure

- The directory is a tree (DIT) of entries; each entry is identified by a **Distinguished Name (DN)**, such as *CN=John Smith,OU=Sales,DC=corp,DC=local*.
- Common attributes: CN (common name), OU (organizational unit), DC (domain component), sAMAccountName, memberOf, mail.

## What enumeration can reveal

If anonymous binds are allowed, or valid low-privilege credentials are obtained, an attacker can pull valid usernames, group memberships, email addresses, computer names, organizational structure, domain policies and service principal names (SPNs). This information supports password spraying and Kerberos-based attacks.

## Tools

- **Active Directory Explorer (AD Explorer)**: Sysinternals GUI AD viewer and editor with snapshots.
- **Nmap NSE**: ldap-rootdse, ldap-search, ldap-brute.
- **ldapsearch** (OpenLDAP), **JXplorer**, **LDAP Admin Tool**, **Softerra LDAP Administrator**, **LDAP Account Manager**, enum4linux.

### Example (authorized lab only)

```
ldapsearch -x -H ldap://10.10.1.22 -s base namingContexts
ldapsearch -x -H ldap://10.10.1.22 -b "dc=corp,dc=local" "(objectClass=user)" sAMAccountName
```

# 5. NFS Enumeration

**Network File System** lets a client mount remote directories as if they were local. Servers publish **exports** (listed in /etc/exports on Unix/Linux). NFS uses **port 2049** and relies on the RPC portmapper on **port 111** and on the mountd service.

Enumeration goals: find the exported directories, which hosts or networks may mount them, and whether the exports are writable or use weak options such as *no_root_squash*, which can let a remote root user act as root on the share.

```
rpcinfo -p 10.10.1.9           # list registered RPC services
showmount -e 10.10.1.9         # list exports
nmap -p 111,2049 --script nfs-ls,nfs-showmount,nfs-statfs 10.10.1.9
```

Other tools: RPCScan, SuperEnum, NFSShell, Nmap NSE nfs-* scripts.

# 6. SMTP and DNS Enumeration

## SMTP enumeration

**Simple Mail Transfer Protocol** (TCP 25, also 465/587) delivers email. Mail servers may reveal whether a mailbox exists through three built-in SMTP commands, letting an attacker build a list of valid users:

| Command | Behaviour |
|---|---|
| VRFY | Verifies whether a given user name or mailbox is valid. |
| EXPN | Expands a mailing list to show all member addresses. |
| RCPT TO | Specifies a recipient; the server's accept/reject response reveals validity. |

```
$ nc -nv 10.10.1.20 25
220 mail.corp.local ESMTP
VRFY alice
250 Alice <alice@corp.local>
VRFY nosuchuser
550 5.1.1 User unknown
```

Tools: Nmap (smtp-enum-users, smtp-commands), smtp-user-enum, Metasploit auxiliary/scanner/smtp/smtp_enum, NetScanTools Pro.

## DNS enumeration

DNS maps names to IP addresses. Enumerating DNS reveals hosts, mail servers, name servers, internal naming conventions and sometimes entire network layouts.

| Record | Meaning |
|---|---|
| A / AAAA | Hostname to IPv4 / IPv6 address |
| MX | Mail exchanger for the domain |
| NS | Authoritative name servers |
| CNAME | Alias pointing to another name |
| SOA | Start of authority: primary server, serial number, refresh timers |

| Record | Meaning |
|---|---|
| PTR | Reverse lookup, IP to hostname |
| SRV | Service location (e.g. domain controllers, SIP) |
| TXT | Free text: SPF, DKIM, verification strings |

### DNS zone transfer

A zone transfer (AXFR) is meant for replication between a primary and secondary server. If a server allows transfers to **any** requester, one query returns the whole zone.

```
nslookup
> set type=NS
> example.com
> server ns1.example.com
> set type=any
> ls -d example.com

dig axfr example.com @ns1.example.com
nmap --script dns-zone-transfer -p 53 ns1.example.com
dnsrecon -d example.com -t axfr
```

### Other DNS techniques

- **Brute-forcing subdomains** with wordlists (dnsrecon, dnsenum, fierce).
- **Reverse lookups** across an IP range.
- **DNS cache snooping**: querying a recursive resolver non-recursively to learn which domains it has recently resolved.
- **DNSSEC zone walking**: following NSEC records to list every name in a signed zone.

## 7. Other Enumeration Techniques

| Technique | What it targets | Notes / tools |
|---|---|---|
| NTP enumeration | UDP 123 | Lists hosts that synchronise with a server. Tools: ntpdate, ntptrace, ntpdc, ntpq, Nmap ntp-info. |
| IPsec enumeration | UDP 500 (IKE) | Detects VPN gateways and the transform sets they accept. Tool: ike-scan. |
| VoIP enumeration | SIP, UDP/TCP 5060 | Discovers extensions, users and phones. Tools: SIPVicious (svmap, svwar), Nmap sip-enum-users. |
| RPC enumeration | TCP/UDP 111, 135 | Lists registered RPC services. Tools: rpcinfo, rpcclient. |
| Telnet enumeration | TCP 23 | Banner grabbing to identify device and OS. |
| FTP / TFTP enumeration | TCP 21 / UDP 69 | Banners, anonymous login, writable directories. Tools: Nmap ftp-anon, tftp. |
| SMB enumeration | TCP 445 | Null sessions, shares, users, policies. Tools: enum4linux, smbclient, Nmap smb-* scripts. |
| IPv6 enumeration | ICMPv6 / NDP | Discovers hosts via multicast. Tools: Nmap -6, Hyena, THC-IPv6, ping6 ff02::1. |
| BGP enumeration | TCP 179 | Reveals routing information and AS numbers. |
| Linux/Unix enumeration | finger, rpcinfo, showmount, rwho | Lists users and exports. Tools: enum4linux, finger, rusers. |

| Technique | What it targets | Notes / tools |
|---|---|---|
| Windows enumeration | Net commands, rpcclient | net user, net localgroup, net accounts, net share, net view. |

### SMB and null sessions

A **null session** is an unauthenticated connection to the IPC$ share. On old or misconfigured Windows systems it allowed listing users, groups and shares. Modern systems restrict this by default, but legacy hosts and some Samba servers may still allow it.

```
net use \\10.10.1.22\IPC$ "" /u:""      (null session, authorized labs only)
enum4linux -a 10.10.1.22
smbclient -L //10.10.1.22 -N
```

#### Windows 'net' commands for enumeration

| Command | Output |
|---|---|
| net view /domain | Domains visible on the network |
| net view \\<host> | Shares offered by the host |
| net user / net user <name> | Local accounts / one account's details |
| net localgroup administrators | Members of the local Administrators group |
| net accounts | Password policy: length, age, lockout |
| net share | Shares on the local machine |

# 8. Enumeration Countermeasures

General principle: expose only what is required, authenticate every query, encrypt traffic, and monitor for unusual enumeration attempts.

### NetBIOS / SMB

- Disable NetBIOS over TCP/IP where it is not needed (adapter settings, WINS tab, or DHCP option).
- Block ports 137, 138, 139 and 445 at the perimeter firewall and between untrusted segments.
- Disable file and printer sharing on hosts that do not need it; disable SMBv1.
- Restrict anonymous access (RestrictAnonymous, Network access policies); remove null session pipes and shares.

### SNMP

- Disable the SNMP service if unused.
- Change default community strings; use long, random strings.
- Migrate to **SNMPv3** with authentication and encryption.
- Restrict SNMP to management hosts by ACL; block UDP 161/162 at the edge.
- Use read-only access where possible and remove write communities.

### LDAP

- Disable anonymous binds; require authentication.
- Enforce LDAP signing and channel binding; use **LDAPS** or STARTTLS.

- Apply least privilege to directory read permissions; restrict who can query sensitive attributes.
- Block 389/636/3268/3269 from untrusted networks; enable auditing of bulk queries.

### NFS and RPC

- Export only required directories to specific hosts; use read-only exports where possible.
- Avoid *no_root_squash*; use root squashing and NFSv4 with Kerberos authentication.
- Block 111 and 2049 at the perimeter; disable unneeded RPC services.

### SMTP

- Disable VRFY and EXPN; configure the server to give uniform responses for valid and invalid recipients.
- Ignore or rate-limit unauthenticated RCPT TO probing; require authentication for relaying.

### DNS

- Restrict zone transfers to named secondary servers (allow-transfer) and secure them with TSIG.
- Use split-horizon DNS so internal records are not visible externally.
- Do not publish host information (HINFO) or sensitive internal names; disable recursion for external clients; use DNSSEC with NSEC3.

### Other services

- NTP: restrict queries with ACLs and disable monlist/mode 7.
- VoIP: authenticate SIP registrations, use SIP over TLS, disable unneeded extensions' responses.
- Telnet/FTP: replace with SSH/SFTP; remove banners; disable anonymous FTP.
- Keep systems patched; harden with baselines; use IDS/IPS to alert on scans and repeated queries.

# 9. Hands-on Labs

The walkthroughs below follow the standard CEH lab layout. IP addresses are examples: replace them with your own lab machines (for instance an attacker VM and a Windows target). Menus and option names can differ slightly between tool versions.

## Lab 1: NetBIOS enumeration using Windows command-line utilities

**Objective:** collect NetBIOS names, sessions and shares of a target using built-in tools.

#### Steps

- Open **Command Prompt** on the attacker Windows machine.
- Run **nbtstat -a** with the target IP or name to view its NetBIOS name table.
- Run **nbtstat -c** to view the local NetBIOS name cache.
- Run **net use** to see current connections and drive mappings.
- Run **net view** to see shares exposed by the target.

```
nbtstat -a 10.10.1.22
nbtstat -A 10.10.1.22
nbtstat -c
net use
net view \\10.10.1.22
```

**Reading the output:** look for <00> UNIQUE (computer name), <00> GROUP (domain/workgroup), <20> UNIQUE (file server running) and <03> UNIQUE (logged-on user). The MAC address is also shown.

## Lab 2: NetBIOS enumeration using NetBIOS Enumerator

**Objective:** scan an IP range graphically for NetBIOS information.

- Install and launch **NetBIOS Enumerator** (by Nir Sofer / SoftPerfect-style utility) on the attacker machine.
- In the IP range field, enter the **start and end IP** of the range to scan (e.g. 10.10.1.5 to 10.10.1.50).
- Click the **Scan** (play) button.
- Wait for results. Hosts that respond appear in the left pane with their name; click a host to see the details such as computer name, user, MAC address and domain/workgroup.
- Expand a host entry to view shared resources, if any are listed.
- Export or screenshot results as evidence in your report.

## Lab 3: NetBIOS enumeration using an Nmap NSE script

**Objective:** use the Nmap Scripting Engine for NetBIOS and SMB information.

```
# NetBIOS name table over UDP 137
nmap -sU -p 137 --script nbstat.nse 10.10.1.22

# SMB / NetBIOS details (TCP 139 and 445)
nmap -p 139,445 --script smb-os-discovery 10.10.1.22
nmap -p 445 --script smb-enum-shares,smb-enum-users 10.10.1.22

# Combined service/version scan with default scripts
nmap -sV -sC -p 137-139,445 10.10.1.22
```

**Output to note:** NetBIOS name, NetBIOS user (usually <unknown> when nobody is logged in), MAC address, OS version, domain and FQDN, share names with access levels. Newer Nmap versions use smb-enum-shares and smb-enum-users; their results depend on authentication.

## Lab 4: SNMP enumeration using snmp-check

**Objective:** extract system, user, network and process information from an SNMP agent. Requirement: the target runs SNMP with a known (or default) community string. **snmp-check** comes preinstalled on Kali / Parrot.

- Confirm SNMP is reachable: *nmap -sU -p 161 10.10.1.22*.
- Run snmp-check against the target, using the default community string (**public**) unless you know another.
- Add **-c** to give a community string, **-v** to choose the SNMP version, and **-w** to test for write access.
- Review the output sections and save it to a file with a redirect.

```
snmp-check 10.10.1.22
snmp-check -c public -v 2c 10.10.1.22
snmp-check -c private -w 10.10.1.22
snmp-check 10.10.1.22 | tee snmp_result.txt
```

| Output section | What it shows |
|---|---|
| System information | Hostname, description, uptime, contact, location |
| User accounts | Local user names |

| Output section | What it shows |
|---|---|
| Network information | IP forwarding, TTL, TCP/UDP stats |
| Network interfaces | Interface names, MAC, speed, traffic counters |
| Network IP / routing | IP addresses, routing table |
| Listening TCP/UDP ports | Open ports and local addresses |
| Processes / Storage / Software | Running processes, disks, installed programs |

Compare with *snmpwalk -v2c -c public 10.10.1.22*, which dumps the raw MIB tree, and Nmap's snmp-info, snmp-processes and snmp-win32-users scripts.

## Lab 5: LDAP enumeration using Active Directory Explorer

**Objective:** browse an Active Directory database using the Sysinternals **AD Explorer** (ADExplorer.exe). It needs network access to a domain controller and a domain account; a standard low-privilege user is usually enough to read most objects.

- Download Sysinternals AD Explorer and run **ADExplorer.exe** (no installation needed); accept the license.
- In the **Connect to Active Directory** dialog, enter the domain controller's **IP address or name** in the Connect to field (port 389 is used).
- Enter the **user name** and **password** of a valid domain account, then click OK.
- Expand the left tree: DC=domain, then CN=Users, CN=Computers, OU=... to view user, group and computer objects.
- Select an object to see its attributes in the right pane (sAMAccountName, memberOf, mail, objectSid, lastLogon, pwdLastSet).
- Use **Search** (Ctrl+F) to query by attribute, and **File > Create Snapshot** to save an offline copy of the directory.
- Use **File > Compare Snapshots** to spot changes between two points in time.

Defender's view: snapshots show exactly what any authenticated user can read. Use them to audit over-permissive attributes and stale accounts.

## Lab 6: NFS enumeration using RPCScan

**Objective:** identify RPC services and NFS exports on a Linux/Unix host. RPCScan is an open-source Python tool that checks for RPC services (rpcinfo-style) and lists NFS exports and their permissions. Target must expose port 111 and/or 2049.

- Confirm the NFS-related ports are open with Nmap.
- Obtain RPCScan (typically by cloning its GitHub repository) and install its Python requirements.
- Run the scan with RPC and mount/NFS checks enabled. Use **--help** to confirm exact options for your version.
- Note the exported paths, allowed clients and any readable or writable shares.

```
nmap -p 111,2049 -sV 10.10.1.9
git clone https://github.com/hegusung/RPCScan.git && cd RPCScan
pip3 install -r requirements.txt
python3 rpc-scan.py 10.10.1.9 --rpc      # enumerate RPC services
python3 rpc-scan.py 10.10.1.9 --mount    # enumerate NFS exports / mounts
python3 rpc-scan.py --help               # verify options for your version
```

Cross-check manually with *showmount -e 10.10.1.9* and *rpcinfo -p 10.10.1.9*. Without network access to GitHub or the target, this lab cannot be run.

### Lab 7: Enumerating information using Global Network Inventory

**Objective:** use **Global Network Inventory**, an agentless network auditing tool for Windows, to collect hardware, software, user and configuration data from machines on a network. It queries remote systems using WMI and related Windows interfaces, so it needs suitable credentials (typically an administrator account) and open ports such as RPC (135) and SMB (445).

- Install and launch **Global Network Inventory**; the Scan Wizard opens (or choose **Scan > New scan**).
- Select the scan mode: **Single address scan**, **IP range scan** or **Domain scan**. For a single machine pick Single address and enter its IP.
- Choose **Authentication**: use the current user or enter the target's administrator credentials.
- Review the scan settings and click **Scan**.
- When the scan completes, open the **Scan results** entry. Browse categories in the left pane.
- Review: BIOS, Bios/System info, Operating system, Installed software and patches, Users and groups, Shares, Services, Processes, Network adapters, Startup commands and Environment variables.
- Export or print a report for documentation.

Related tools: SoftPerfect Network Scanner, Advanced IP Scanner, PRTG, SolarWinds Network Performance Monitor, Nsauditor, Hyena, Lansweeper.

# 10. Quick Revision Sheet

| Topic | Remember |
|---|---|
| Enumeration | Active connection + directed queries; follows scanning; extracts users, shares, services. |
| NetBIOS | UDP 137/138, TCP 139; 16-char names; <20> file server, <03> messenger/user, <1C> DC; nbtstat -a / -A / -c. |
| SNMP | UDP 161/162; community strings public / private; v3 secure; MIB and OID; snmp-check, snmpwalk. |
| LDAP | TCP 389, LDAPS 636, GC 3268; DN, CN, OU, DC; anonymous bind risk; AD Explorer, ldapsearch. |
| NFS | Port 2049 + RPC 111; showmount -e; no_root_squash risk; RPCScan. |
| SMTP | Port 25; VRFY, EXPN, RCPT TO; smtp-user-enum. |
| DNS | Port 53; zone transfer AXFR over TCP; dig axfr, nslookup ls -d; restrict transfers. |
| NTP / IPsec / VoIP | 123 / 500 / 5060; ntpdc, ike-scan, SIPVicious. |
| Countermeasures | Disable unused services, change defaults, SNMPv3, LDAPS + signing, restrict zone transfers, firewall ports, patch. |

# 11. Practice Questions

| # | Question | Answer |
|---|---|---|
| 1 | Which NetBIOS suffix indicates that the File Server service is running? | <20> UNIQUE. |

| # | Question | Answer |
|---|---|---|
| 2 | Which port does SNMP use for trap messages? | UDP 162 (queries use UDP 161). |
| 3 | What are the default SNMP read-only and read-write community strings? | public and private. |
| 4 | Which SMTP commands can reveal valid users? | VRFY, EXPN and RCPT TO. |
| 5 | What DNS record type lists the mail servers of a domain? | MX. |
| 6 | Which SNMP version adds authentication and encryption? | SNMPv3. |
| 7 | Which LDAP port is used for encrypted connections? | TCP 636 (LDAPS); the Global Catalog equivalent is 3269. |
| 8 | Name the Windows built-in command that displays a remote NetBIOS name table. | nbtstat -a (by name) or nbtstat -A (by IP). |
| 9 | Which Nmap script retrieves NetBIOS names via UDP 137? | nbstat.nse. |
| 10 | Which command lists NFS exports on a server? | showmount -e <server>. |
| 11 | What is the best defence against DNS zone transfer abuse? | Allow transfers only to authorized secondary servers (and use TSIG). |
| 12 | What is Sysinternals AD Explorer's snapshot feature used for? | Saving an offline copy of the AD database and comparing snapshots to detect changes. |

Good luck with your studies. Practice every lab in an isolated, authorized environment and always keep screenshots and notes of the commands and outputs.